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Laser phase plate improves structure determination of small proteins by cryo-EM

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Phase plates can in principle overcome the poor image contrast in electron cryo-microscopy (cryo-EM) and the resulting limits on the structural reconstruction of small proteins. However, previous designs have been unstable and compromised the high-resolution signal. They have thus been unable to surpass results achieved by standard cryo-EM. Here, we show that the laser phase plate (LPP), installed in a custom, modern Titan Krios microscope, enhances the resolution in single-particle reconstruction of small proteins by improving specimen-motion correction, recovery of information from the early frames, as well as particle visualization, 3D classification, and alignment. These advances use standard defocus ranges and reconstruction procedures, but open the door to LPP-tailored protocols offering further improvements by leveraging the LPP demonstrated here.

Electron cryo-microscopy (cryo-EM) has already had revolutionary impacts on cell and molecular biology and has become a major source of structural information (1, 2). However, its performance remains far short of its potential: While it should be possible to reconstruct particles as small as 17 kDa (3) or even smaller (4), in practice proteins < 50 kDa remain difficult to reconstruct, and those below 100 kDa constitute only about 10% of cryo-EM structures deposited in the Electron Microscopy Data Bank (EMDB) in 2025 (5) despite making up ~90% of the human proteome (6). Moreover, data from many thousands of particles must be merged to obtain < 3.5 Å resolution; this makes structure determination laborious and hampers analysis of heterogeneous particles (4, 7).

The root of the problem is that biological specimens are weak-phase objects: They only apply a small phase shift to the electron wave function, but they do not absorb electrons appreciably. As a result, a perfect image has nearly zero contrast. Intentional defocus is used to partially convert phase shifts into image contrast, but this conversion is ineffective for the low-spatial-frequency components of an image, which are often important for identifying and orienting particles. Moreover, zeros in the resulting contrast transfer function (CTF) allow only 50%, at best, of the potential high-spatial-frequency information to be retained.

It was pointed out long ago (8, 9) that optimal image contrast can be achieved with phase plates, which apply a 90° phase shift to the undiffracted part of the electron wave function (Fig. 1). Even the widely-adopted Volta phase plate, however, shows instability (10), an ~18% signal loss (11), and additional unexplained loss at high spatial frequencies (12). After promising initial results,

a careful study of single-particle analysis (SPA), using the 167-kDa PAC1 GPCR receptor complex, found that using the Volta phase plate made the final resolution slightly worse in spite of increased low-spatial-frequency contrast (13).

We have developed a laser phase plate (LPP) and demonstrated that it has superior stability while also showing negligible signal attenuation (14–17) and decoherence from thermal effects (18). Instead of placing a material phase shifter into the beam path, it induces the phase shift by stimulated Compton scattering (19) with a continuous light intensity of ~400 GW/cm², the highest ever demonstrated. A 12-W, 1064-nm laser beam is built up to 80 kW circulating power in a Fabry-Pérot cavity, where it is focused to a waist of 6.7 μm (1/e² intensity radius) (16). Initial experiments tested the LPP within an older-generation electron microscope which was limited by a large beam-energy spread, high aberration coefficients, and an older-generation specimen stage and camera. As a result, no phase plate has demonstrated a resolution improvement in SPA to date (13, 17).

We now report the use of the LPP inside a modern electron microscope (Thermo Fisher Krios G4 with a CEOS CETCOR spherical aberration corrector, Fig. 1A) and demonstrate clear, quantitative improvements in SPA with small proteins. We compared laser-on and laser-off imaging with three specimens each of rabbit muscle aldolase (157 kDa) and human hemoglobin (64 kDa). Images were recorded under similar conditions, and structures were determined using standard RELION (20) processing.

In all six cases, the LPP improved the resolution that was achieved when using nearly-identical, relatively small datasets (Fig. 1H). The improvement was greater for hemoglobin, the

smaller (thus more challenging) of the two proteins, which yielded the poorest resolution without the LPP. Since we used standard data acquisition parameters (including defocus) for this comparison, we do not expect the LPP to improve the high-spatial-frequency (herein “HSF,” smaller features than $\sim[10 \text{ \AA}]^{-1}$) information (fig. S1). Rather, as we show here, the improved low-spatial-frequency [herein “LSF,” larger features than $\sim[10 \text{ \AA}]^{-1}$] contrast improves motion correction of micrographs, identification and orientation of particles, and utilization of information from the early frames (which have suffered the least radiation damage), culminating in superior resolution. Better correction of specimen motion and improved initial classification is observed in all data with the LPP, even in datasets where a strong improvement in the final resolution is not obtained. We believe these results open the door to further development, including reconstruction of particles that are currently considered to be too small or too structurally variable, and a strong improvement in the ability to identify small particles in tomograms.

Experimental results

Aldolase and hemoglobin were chosen as test specimens because they are large enough for successful reconstruction using conventional cryo-EM but small enough that results can be expected to depend critically on imaging performance. At 157 kDa, aldolase is comparable in size to the complex used in Ref. (13) for quantitative characterization of the Volta phase plate (167 kDa) and frequently used for benchmarking (21–23). At 64 kDa, hemoglobin is on the small end of proteins studied by cryo-EM, with less than 5% of structures released on EMDB in 2025 being smaller (5), but has been previously reconstructed both with the Volta phase plate (24) and without (23, 25–27). Both specimens are straightforward to prepare by conventional plunge-freezing. In six experiments, termed A1–3 (aldolase) and H1–3 (hemoglobin), we collected pairs of datasets with the LPP turned on and off. The symmetry of the proteins, D_2 for aldolase and C_2 for hemoglobin, provides multiple asymmetric units per particle, which facilitates high-resolution reconstruction from small datasets, consisting, in the current work, of only ~ 9000 to $\sim 26,000$ particles extracted from ~ 100 to 250 micrographs.

Keeping the imaging parameters and analysis procedure as similar as possible between laser-on and -off allows us to trace the role of the LPP through the reconstruction pipeline. In each experiment, laser-on and -off data were collected from the same grid, with similar numbers of micrographs and a similar distribution of defocus values (see table S1) (13, 28). The microscope was aligned under laser-off conditions; laser-on data collection then required slightly adjusting the electron beam deflectors to intersect the laser and electron beams (17). The resulting few- μm translation of the electron beam somewhat increased the optical aberrations in laser-on data, but not in laser-off data. Other than that, the configuration of the microscope electron optics was

exactly the same for laser on and laser-off data acquisition (Materials and Methods). In both configurations, chromatic aberration (C_c) was 5.3 mm and spherical aberration was ~ 0.01 mm.

To control for variations in specimen quality throughout the grid, we imaged the same grid squares in experiments A2 to A3 and H2 to H3, dividing the holes in each square between the laser-on and laser-off conditions. Although we have not observed a substantial ice contamination rate in our microscope, we either collected laser-off data first (A2, H2) or interleaved data collection so that half of the squares were first imaged with the LPP on and the other half with it off (A3, H3). Only in experiments A1 and H1, each grid square was imaged with a single condition (comparable squares having been selected manually prior to collection) and laser-on data were collected first. Although results from the first round may be affected to some extent by this difference, we include them for completeness.

We measured the final resolution via the gold-standard Fourier shell correlation (FSC) using RELION (Fig 1, F and H, and Fig. 2). The LPP yielded the strongest improvements in resolution (the largest being to 3.09 $\text{\AA}$ from 4.46 $\text{\AA}$) for the hemoglobin specimens where the laser-off resolution was low. Even the worst laser-on hemoglobin resolution obtained was 0.25 $\text{\AA}$ better than the best one with the laser off. The aldolase specimens (the larger and more symmetric of the two particles), which already give good resolution without the laser, exhibit a smaller laser-on advantage of between 0.03 $\text{\AA}$ and 0.47 $\text{\AA}$ (Fig. 2).

Effect of the LPP on reconstruction

To illustrate the effect of the LPP at different stages of the reconstruction process, we now describe in detail the results obtained in experiment H2. These data produced the best resolution for both laser-on and laser-off among our hemoglobin experiments, enabling robust analysis especially of HSF information. This experiment showed a large (but not the largest) resolution improvement from the LPP (0.48 $\text{\AA}$), despite experimental conditions slightly favoring the laser-off data as described above. The results are summarized in Fig. 1, F and G, and indicated in red in Fig. 1H. The reconstruction workflow is summarized in fig. S2.

Laser phase contrast readily visualizes small proteins

Although imaging without defocus would be optimal when using the LPP, as that would avoid oscillations in the CTF at high spatial frequencies, there is no established method for doing this reliably. Instead, in these experiments we used a standard defocus of several hundred nanometers (e.g., fig. S1). This allows for CTF estimation with standard software (29) while still providing the laser-off images with enough LSF signal to proceed with particle identification and reconstruction. It also keeps the overall HSF signal strength similar in the laser-on and -off datasets.

The LPP makes it much easier to identify by eye not only the presence of particles (e.g., Fig. 1, B and C, and fig. S3) but also structural variations between particles as well as variations within the ice, which are difficult or impossible to distinguish without using a phase plate (17). This makes evaluating specimen quality during imaging easier.

With the defocus setting used here, automated reference-based particle picking with RELION was able to detect the hemoglobin particles in both laser-on and laser-off data. The threshold for particle detection was straightforward to set in laser-on data. Picking was more error-prone in laser-off data, with more aggressive thresholding needed to achieve a similar detection density. Ultimately, in experiment H2, automated picking identified a similar number of candidate particles (>300,000) in each dataset from a similar number of micrographs.

Laser phase contrast improves motion correction

Specimen motion correction, the first step of any data processing pipeline, is improved in movies acquired with the LPP. Data were recorded in electron event representation (EER) format (30). Binning the exposures into frames of $1 \text{ e}/\text{\AA}^2$, we performed RELION's patch motion correction (31) to track both global and local motion in a 5×5 grid within each micrograph. While global specimen motion is relatively large during the first few frames of a movie due to our choice of specimen support (materials and methods), it eventually equilibrates and the lengths of frame-to-frame displacements stabilize (Fig. 3A). Global displacements in the stable regime, when aggregated over a dataset, are well-described by a single Rayleigh distribution characterized by a scale factor, σ (32). Both the true motion and noise of displacement estimation contribute to the apparent specimen motion. Since true specimen motion is independent of the laser configuration, we deduce that the decrease in σ to $0.210 \text{ \AA}$ (on) from $0.264 \text{ \AA}$ (off) reflects better global motion correction in LPP data (Fig. 3A, inset).

We also find that the LPP makes patch motion correction more robust. We compared the raw frame-to-frame shift in each patch to the estimate provided by its low-order polynomial approximation (33, 34). We evaluated the polynomial at the centroid of each patch and in each frame. Averaging over all patches and micrographs, the difference between the raw and smoothed trajectories has a standard deviation lower by a factor of 1.5 to 5 throughout the movie when the LPP is on, with laser-off trajectory estimates gradually worsening throughout the movie. The performance of laser-on data are both better and essentially dose-independent (Fig. 3B).

Each of the six laser-on datasets showed better motion correction than its laser-off counterpart, even when there was little improvement in the final resolution (Fig. 3C). This suggests that, while decidedly advantageous, improved motion tracking is not on its own responsible for the resolution improvements that the LPP yields. Per-frame *B*-factor plots obtained via Bayesian

polishing in RELION only show a clear improvement in a subset of the experiments (fig. S4).

Better initial 3D classification is achieved with laser phase contrast

In all six experiments, laser-on data achieved better initial classification than without the LPP. After extracting the initial set of particles from the micrographs, reconstructions proceeded with a three-dimensional (3D) classification to begin removing low-quality particles and spurious picks. In experiment H2, the initial 3D classification eliminated 75% of the laser-off and 70% of the laser-on particles, leaving 88,000 and 97,000, respectively. Reconstructions in the other experiments behaved similarly. To characterize the quality of this initial classification step, we performed 3D refinement in RELION using the resulting particle stacks, obtaining per-particle rotational and translational alignments and a resulting structure. Laser-on data produced a structure with better estimated resolution ($3.64 \text{ \AA}$) than laser-off data ($4.30 \text{ \AA}$), see Fig. 3, D and E. We found this to be a general result (Fig. 3F). Even for aldolase experiments A2 and A3, in which the resolution gap between laser-on and -off structures decreases by the end of the reconstruction process, we still observe substantial resolution improvements (0.14 and $0.44 \text{ \AA}$, respectively) from the LPP in the first classification step.

The LPP increases low-spatial-frequency signal that is used during particle alignment

Using the final particle stacks from experiment H2, we performed 3D refinement in RELION with a high-pass filter (HPF) applied during the expectation steps of the iterative procedure (Materials and Methods). As shown in Fig. 4A, the HPF has no effect on final reconstruction resolution reported by RELION when the cut-on frequency is below $\sim 0.025 \text{ \AA}^{-1}$. The signal omitted in that case may aid detection of hemoglobin particles (diameter $\sim 65 \text{ \AA}$) (35), but by this stage in the reconstruction it is no longer useful. However, in the regime between $\sim 1/(40 \text{ \AA})$ and $\sim 1/(18.5 \text{ \AA})$ (shaded in the figure) the resolution of the laser-on refinement gradually deteriorates, while that of the laser-off refinement remains largely unaffected. For cut-on frequencies greater than $1/(18.5 \text{ \AA})$, the resolution remains better in the laser-on case by $\sim 0.2 \text{ \AA}$, reduced from the gap of $\sim 0.5 \text{ \AA}$ without filtering. We note that the spatial frequency band we identify agrees well with the band $1/(30 \text{ \AA})$ to $1/(10 \text{ \AA})$ determined by Henderson and McMullan to be important for orientation estimation (36). When the HPF cut-on is increased above $0.06 \text{ \AA}^{-1}$, too little LSF signal remains and alignment fails. In short, a sufficiently aggressive high-pass filter nearly equalizes the resolution achieved in the two datasets, mostly by excluding the extra LSF signal captured when using the LPP. This shows that LSF signal aids particle alignment, partly explaining the resolution improvements observed with the LPP.

The LPP enables greater utilization of high-resolution signal from early frames

While LSF information appears to aid 3D refinement, HSF information is ultimately needed to achieve higher resolution. Since HSF information is much more rapidly degraded by radiation-induced damage than LSF information (37), it is predominantly contained in the early frames of a micrograph. We therefore looked at the effect of omitting the initial movie frames on the reconstructions in experiment H2 (Fig. 4B) (33). We found that whereas the first five frames (with 1 e/Å²/frame) did little to improve the resolution in laser-off data (to 3.35 Å from 3.47 Å), they improved the laser-on reconstruction resolution substantially (to 2.87 Å from 3.30 Å). This demonstrates that HSF signal, which is nominally of similar strength in both datasets based on the CTF (e.g., fig. S1), is incorporated into the reconstruction more effectively when we use the LPP. Omission of additional frames beyond the first five confers a different, lower per-frame resolution loss in laser-on data, a feature seemingly not observed in laser-off data. Since the early frames contain much more specimen motion (Fig. 3A), we speculate that this piecewise behavior is due in part to improved motion correction.

Discussion and outlook

We have demonstrated several benefits of using the LPP, even when using conditions comparable to those used in standard cryo-EM imaging, even though this compromises the HSF information that could in principle be recovered by imaging in-focus (38). We find the LPP improves the resolution of challenging specimens, with the advantage of using the LPP increasing as the benchmark (laser-off) resolution worsens. Our experiments demonstrate that advantages over conventional cryo-EM can already be extracted solely from the LPP's increase of the LSF portion of the CTF. The identified improvements to SPA results were robust across all six of our experiments, even when the final resolution achieved without the LPP was comparable. The large variation observed between experiments illustrates that good specimen preparation is nevertheless critical to high performance (fig. S5).

The demonstrated improvements in the precision of both global and local motion correction are expected to be of further significance in cryo-electron tomography, which is often limited by the larger motions induced in tilted specimens and the low per-frame fluence (39). It remains to be seen whether LPP-enhanced contrast permits superior results from finer binning of EER movies (<1 e/Å²/frame).

The observed superior results with the LPP in the initial classification of detected particles could lead to shorter SPA workflows when targeting an intermediate resolution, or to better triaging of datasets before investing time in a full processing effort.

We also demonstrated that the additional LSF signal obtained with the LPP helps in better utilization of the available HSF

information: By applying high-pass filters during 3D refinement, we determined that the enhanced LSF signal contributes appreciably to particle alignment at high resolution. We also observed that the LPP enables more effective use of the data during the first ~5 e/Å² of exposure, which contains the most HSF signal but is usually less effectively used because of its strong specimen motion. It is expected, of course, that using holey gold grids (40) and smaller hole sizes (41) will make improving motion correction less important. In future work, this can be quantified, but to do so will require developing an imaging protocol compatible with laser-electron beam alignment (Materials and Methods).

The behavior of the LPP in these experiments is close to the ideal of a stable 90° phase shift, with our most recent round of experiments having a phase shift of $88 \pm 7^\circ$ (see fig. S6 and table S1). However, laser-on imaging can be improved further. Mechanical misalignment by a few micrometers between the LPP focus and the transfer lens rotation center necessitated deflecting the electron beam by a few micrometers in the diffraction plane, changing electron propagation through the corrector. This introduces slight optical aberrations in laser-on mode which could not be compensated easily during imaging and were instead addressed computationally (Materials and Methods). Furthermore, specialized processing which incorporates the shape of the laser standing wave in the CTF model is still in development. Finally, throughput in these experiments was limited to ~1 micrograph per minute due to translating the stage to each grid hole. Accelerating data collection, such as by implementing beam-image shift acquisition (42) in tandem with laser-electron alignment, will empower exploration of specimens not previously reconstructed.

In the future, the benefits of increasing the LSF, demonstrated in this study, might be compounded with those resulting from the addition of further HSF information, which becomes available when imaging in-focus. The latter would double the HSF signal power by eliminating oscillations of the CTF (fig. S1) and could be combined with chromatic aberration correction to incorporate additional signal from inelastically-scattered electrons (43). In the case of cryo-EM, however, this long-sought goal requires further development of low-dose, in-focus image acquisition and CTF estimation methods, which currently do not exist. It is anticipated that achieving this goal will bring the molecular weight threshold accessible by SPA below what has been achieved without the LPP (44–46), opening the field of cryo-EM to new targets representing a large swath of the proteome.

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S6

Table S1

Data S1

References (47–51)

MDAR Reproducibility Checklist

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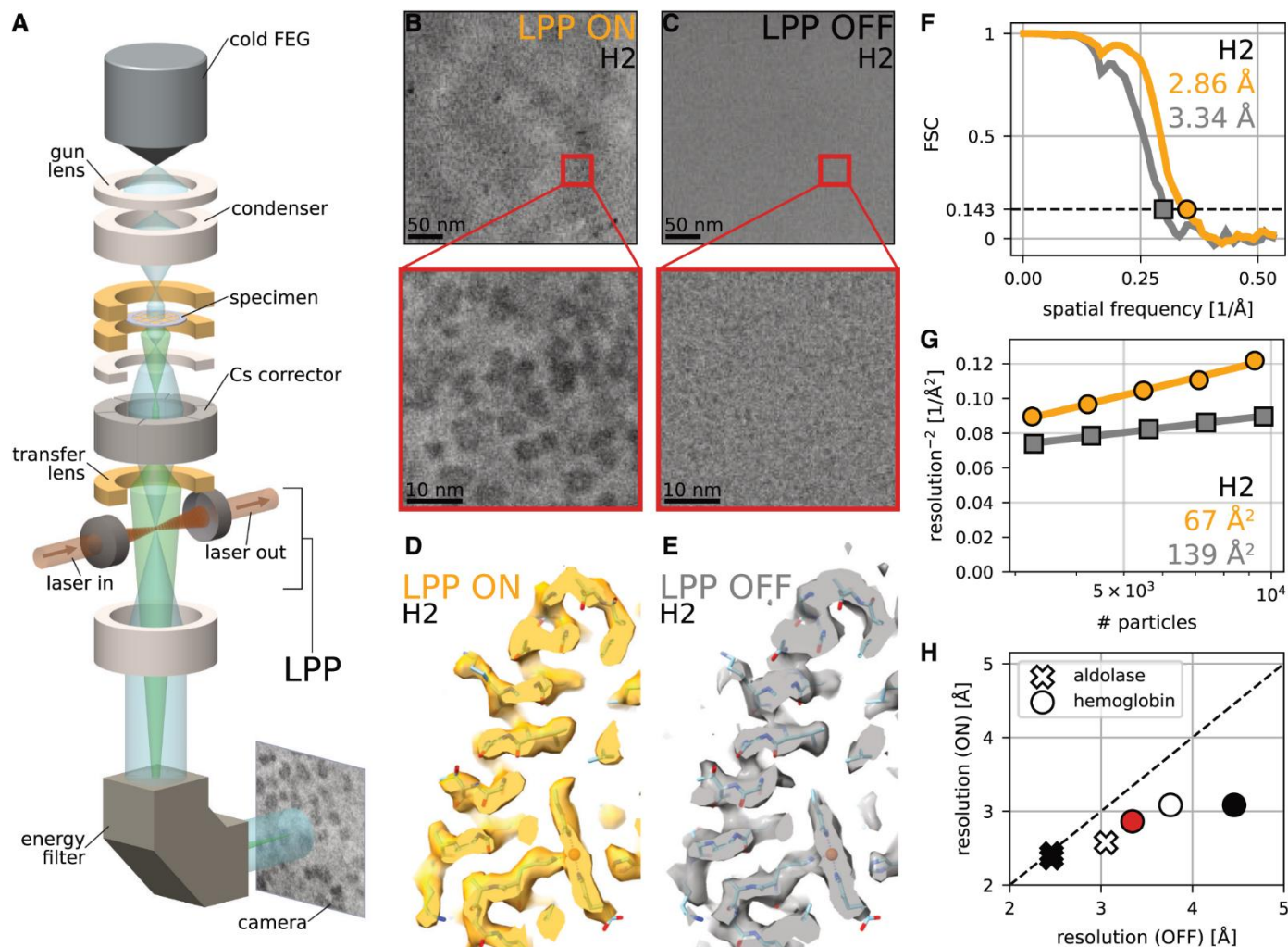


Fig. 1. Laser phase contrast. (A) Schematic of the LPP-Krios, featuring a cold field-emission gun (C-FEG), a spherical aberration (Cs) corrector, an energy filter, and a Falcon 4i direct-detection camera. A transfer lens generates a diffraction plane, where the focus of the LPP laser (brown) intersects the focused unscattered electron beam (blue). (B to C) Example hemoglobin micrographs with LPP on (B) and off (C) from experiment H2 (see text). Each is motion-corrected, divided by its mean, and plotted with the color scale [0.8 (black), 1.2 (white)]. (D to E) Thresholded portions of the final structures, docked with an atomic model (PDB 6NBC) (25). (F) Gold-standard FSC curves showing resolution enhancement to 2.86 Å (on) from 3.34 Å (off). (G) Rosenthal-Henderson plots showing the enhancement *B*-factor to 67 Å² (on) from 139 Å² (off). (H) Comparison of resolutions obtained with the LPP off and on in all six experiments. White markers indicate experiments A1 and H1, red indicates experiment H2, and black indicates the remaining three experiments.

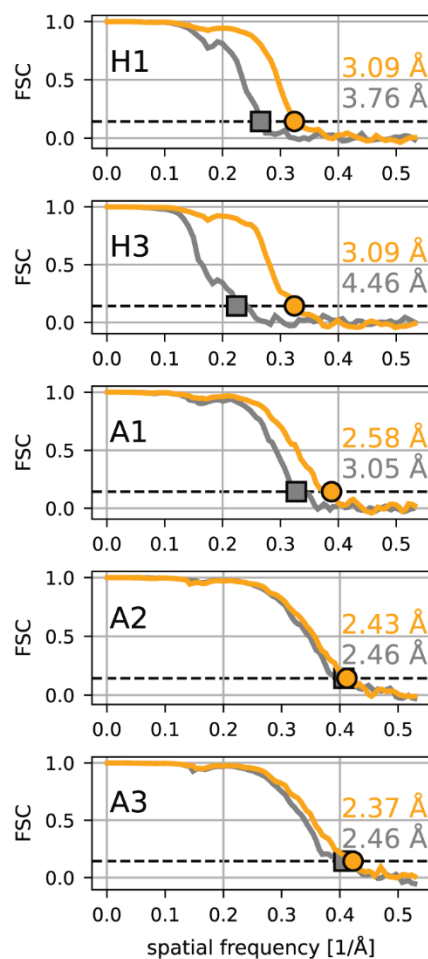


Fig. 2. LPP improves reconstruction resolution. Gold-standard Fourier shell correlation (FSC) plots and final resolution estimates for the five experiments not shown in Fig. 1, with laser-on in orange and laser-off in gray. Dashed black lines indicate a FSC value of 0.143; markers indicate intersection of the FSC curves with this line.

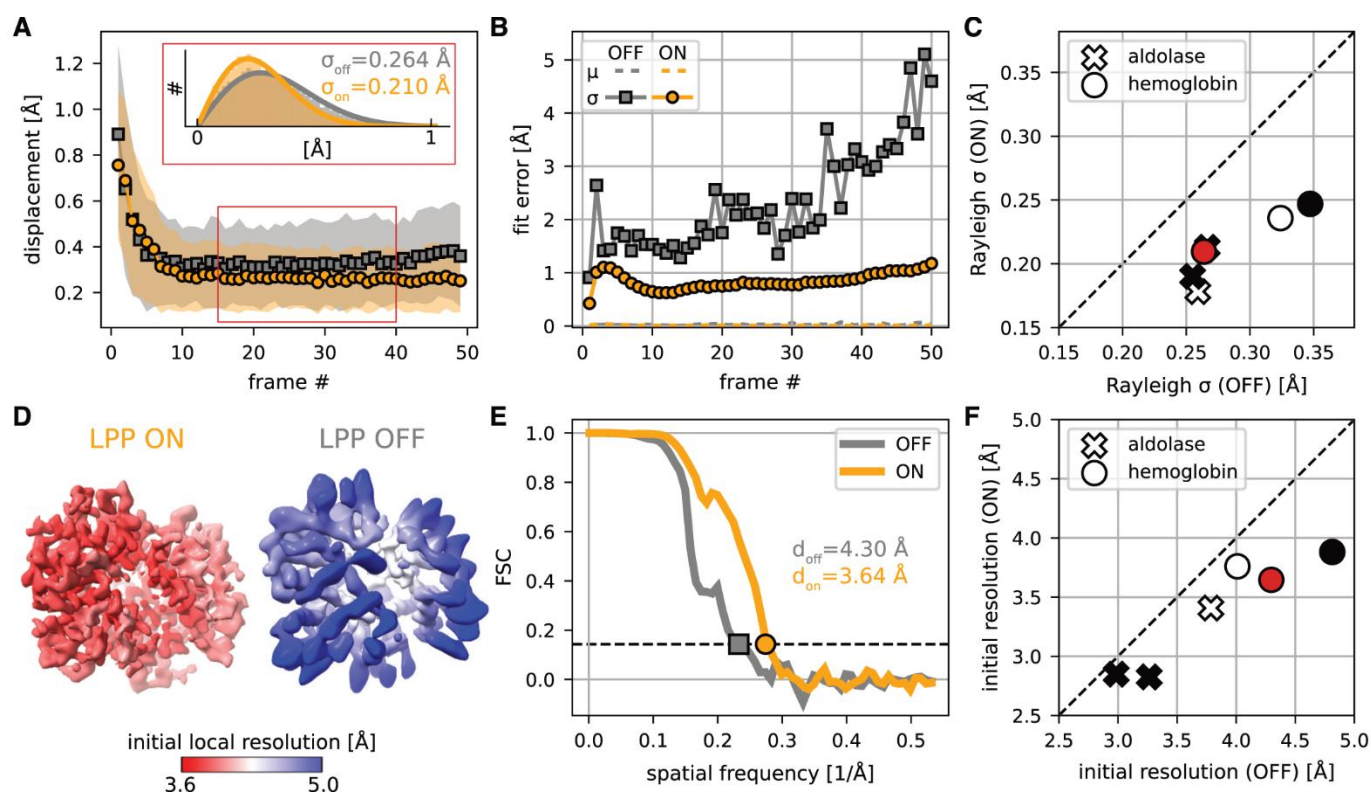


Fig. 3. LPP improves early stages of reconstruction. (A) Displacement between frames over time shows similar overall global motion but a lower value at long times with LPP on, indicating superior precision. Inset shows a histogram of displacements between frames 15 and 40 fit to a Rayleigh distribution. (B) Comparing measured local motion to the low-order polynomial fit determined by RELION's motion correction shows no net bias μ in either case (dashed lines) but substantially better and less time-dependent precision σ of local motion correction with the LPP on (solid lines with markers). Each data point in (A) and (B) is averaged over all micrographs within the dataset and shaded regions in (A) indicate standard deviation. Frames contain $1 \text{ e}/\text{Å}^2$. (C) Rayleigh σ fits compared between LPP on and off. (D) First reconstructed models after initial classification, colored by local resolution. (E) Gold-standard FSC curves for the structures in (D). The dashed line indicates FSC of 0.143. (F) Comparison of resolutions obtained from FSC curves after initial classification with laser off and on for all six experiments. Panels (A), (B), and (D) to (E) show results from experiment H2.

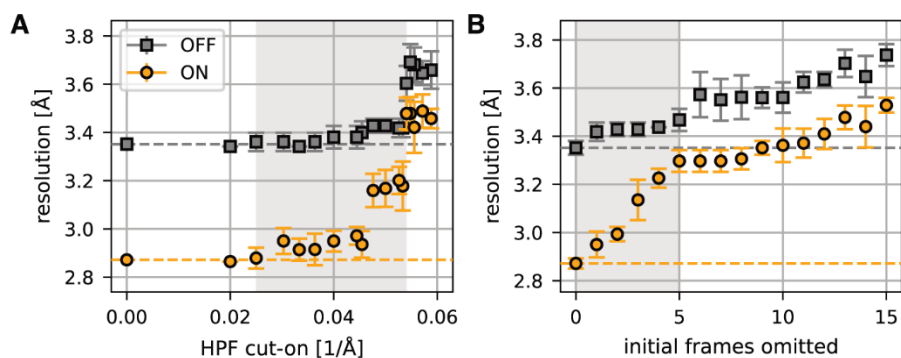


Fig. 4. Characterizing resolution improvement by the LPP. (A) Cut-on frequency of a high-pass filter (HPF) applied during the expectation steps of the final 3D refinement procedure affects resolution. Behavior of laser-on data between cut-on frequencies of $1/(40 \text{ Å})$ and $1/(18.5 \text{ Å})$ (shaded region) indicates contributions from LSF signal to refinement. (B) Omitting initial frames ($1 \text{ e/Å}^2/\text{frame}$) during the final refinement procedure degrades resolution. The shaded region shows that signal from early frames widens the resolution gap between laser-on and laser-off data. In both panels, markers indicate the mean of 10 tests, error bars indicate standard deviation, and dashed lines indicate resolution values obtained with no filter and all frames included. Orange circles correspond to laser-on and gray squares to laser-off. All data are from experiment H2.



Laser phase plate improves structure determination of small proteins by cryo-EM

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